

Authored by: Arpit Raj, LNMIIT Jaipur

LeetCode 459 - Repeated Substring Pattern

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The following C++ solution elegantly verifies whether a given string can be constructed by taking a substring and appending multiple copies of it together. The code relies on systematically testing possible substring lengths from 1 up to half the string's length.

```
class Solution {
public:
    bool repeatedSubstringPattern(string s) {

        int n = s.size();

        for(int p = 1; p <= n / 2; p++) {
            if(n % p != 0) continue;

            string pattern = s.substr(0, p);
            bool ans = true;

            for(int i = p; i < n; i += p) {
                if(s.substr(i, p) != pattern) {
                    ans = false;
                    break;
                }
            }

            if(ans) return true;
        }

        return false;
    }
};
```